

Causal Foundation Model for Decision Makers

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1 Introduction

Causal Foundation Models (CFMs) such as Do-PFN [1], “Use What You Know” [2], MACE-TNP [3] and CausalPFN [4] amortise Bayesian inference over structural causal models (SCMs) using the prior-data-fitted networks paradigm [5, 6]. For an individual with covariates x and an observational dataset D_{obs} , they produce the posterior predictive

$$q_{\theta}(Y \mid D_{\text{obs}}, x, do(t))$$

for each intervention value t . The Conditional Average Treatment Effect (CATE) and Average Treatment Effect (ATE) are then reported as scalar estimates by taking posterior means and subtracting.

This document argues that the scalar-summary output of current CFMs is mismatched with the needs of a decision-maker, and proposes modifications that turn a CFM into a tool that reports an explicit distribution over causal effects. We refer to the resulting model as a *Causal Foundation Model for Decision Makers*. We will incorporate partial-graph conditioning following the “Use What You Know” formulation, so that any domain knowledge available at test time can sharpen the reported distribution.

Why the treatment effect, not the outcome. The prior-data-fitted-network paradigm originates with TabPFN [6], where the object of interest is the outcome Y itself: the task is prediction, and the predictive distribution over Y is the final deliverable. Current CFMs inherit this design — they output a posterior predictive over Y under each intervention and treat that as the primary object. But in causal inference the quantity a decision-maker acts on is not Y ; it is the treatment effect $\tau(x) = Y^{do(1)} - Y^{do(0)}$, and its population average. The outcome distribution under a single intervention is, on its own, of secondary interest: it answers “what will happen under treatment t ,” not “what is the effect of treatment.” Inheriting the TabPFN output design therefore leaves CFMs reporting a distribution over the wrong quantity, and recovering the right one — the distribution of τ — only implicitly and lossily, through a difference of posterior means. We argue that a causal foundation model should output the distribution of the causal effect directly, and should treat the per-intervention outcome distributions as intermediate quantities rather than as the deliverable.

Why minimizing the error could be suboptimal. Reporting the posterior mean of $\tau(x)$ is the choice that minimises total squared error across many independent decisions: if a practitioner must make a thousand causal judgements, returning the mean each time minimises aggregate error over the thousand. But this is an argument about long-run average performance, not about any

single decision. In any individual case the mean can be a poor summary — in particular when the model’s posterior over causal mechanisms is diffuse, uniform, or multimodal, so that the treatment effect could plausibly arise from one of several distinct causal worlds. Collapsing that structure to a single number hides exactly the information a decision-maker needs. Providing the full distribution of τ lets the decision-maker see the competing possibilities and judge how to proceed, rather than acting on a summary that may be suboptimal for the case in front of them.

Contributions. Building on this position, we propose a CFM for Decision Makers that: (i) replaces the two independent per-intervention outcome heads with a *paired potential-outcome head* that outputs the joint distribution of $(Y^{do(1)}, Y^{do(0)})$, preserving the coupling required to compute the treatment-effect distribution without spurious artefacts; (ii) implements this head as a two-dimensional bar distribution on a $J \times J$ grid with $J \ll K$, so the joint head remains computationally light; (iii) applies bivariate DensOLog [8] density refinement to the coarse joint output, recovering a continuous joint density — with marginal potential-outcome densities and the CATE density obtained from it, and treatment-effect moments following from the joint moments of its EM procedure; (iv) aggregates pointwise CATE distributions into an ATE distribution via a dependence-preserving shared-rank coupling; and (v) reweights the synthetic training prior toward domain-relevant causal structures using an LLM-guided sampler.

2 Motivating example and the conceptual gap

Sometimes multimodality can arise in the *model’s* posterior, because the observational distribution is consistent with more than one SCM. The aim of a decision-friendly CFM is to convey this epistemic uncertainty.

Counter-example. Consider a synthetic case in which two SCMs ψ_a, ψ_b are both consistent with D_{obs} in the model’s prior, and the (implicit) PFN posterior places mass roughly $\frac{1}{2}$ on each. Suppose that, for some query x , the two SCMs imply pairs of potential outcomes

$$(Y^{do(1)}, Y^{do(0)})_{\psi_a} = (10, 5), \quad (Y^{do(1)}, Y^{do(0)})_{\psi_b} = (2, 1).$$

Under ψ_a the individual treatment effect is $\tau_a = 5$; under ψ_b it is $\tau_b = 1$.

Current pipeline. The two marginal predictives have mass at $\{10, 2\}$ and $\{5, 1\}$ respectively. The reported CATE is $\hat{\tau}(x) = \frac{1}{2}(10 + 2) - \frac{1}{2}(5 + 1) = 3$. This single number gives the decision-maker no indication that the model is internally torn between $\tau = 5$ and $\tau = 1$.

Naive recovery is also wrong. If one tried to recover the τ -distribution from the two marginals by treating them as independent (e.g. convolving them), one would obtain mass at

$$\{10 - 5, 10 - 1, 2 - 5, 2 - 1\} = \{5, 9, -3, 1\}.$$

The values 9 and -3 are *spurious*: they pair ψ_a ’s treated outcome with ψ_b ’s control outcome and vice versa, which corresponds to no causal mechanism the model itself considers plausible. The marginals do not determine the joint, so any aggregation that ignores the coupling between $Y^{do(1)}$ and $Y^{do(0)}$ can fabricate causal scenarios that the model does not entertain.

Diagnosis. The current architecture performs two *independent* forward passes (one per intervention), discarding the coupling between $Y^{do(1)}$ and $Y^{do(0)}$. The downstream reduction to a scalar then collapses whatever epistemic uncertainty remains in each marginal. Figure 1 illustrates this end-to-end: two independent queries yield two predictive distributions, which are reduced to a single number, leaving the decision-maker without a useful summary of model uncertainty.

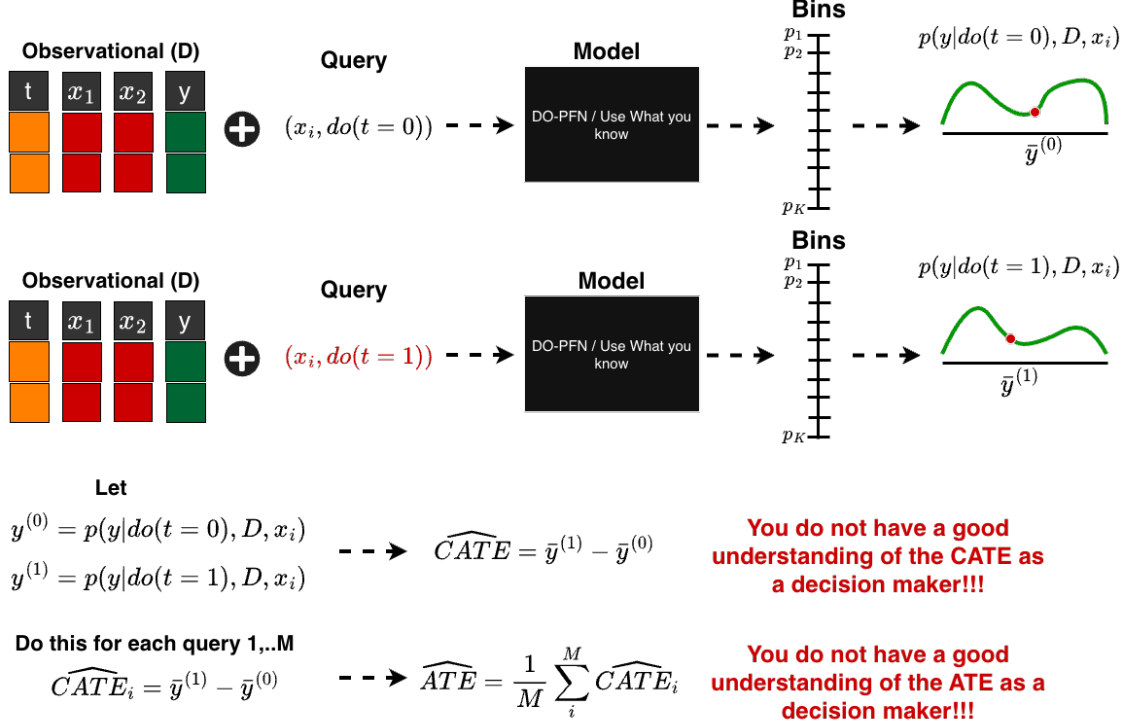


Figure 1: The current pipeline. Two queries (one per intervention) yield two predictive distributions for Y , which can be multimodal whenever the model is uncertain about the underlying causal structure. These two marginals are reduced to a scalar CATE estimate by subtracting posterior means, and aggregated to a scalar ATE. The joint coupling between $Y^{do(1)}$ and $Y^{do(0)}$ is never represented, and the decision-maker only sees the scalar.

3 Background and setup

Let $D_{\text{obs}} = \{(t_i, x_i, y_i)\}_{i=1}^n$ denote an observational dataset of factual outcomes, with treatment $t \in \{0, 1\}$, covariates $x \in \mathbb{R}^d$, and outcome $y \in \mathbb{R}$. The two potential outcomes are written $Y^{do(0)}$ and $Y^{do(1)}$, and the treatment effect is $\tau(x) = Y^{do(1)} - Y^{do(0)}$.

Existing CFMs implement the output head as a bar distribution [6] with K uniform-width bins over a normalised range of Y (typically $K = 1000$). Training proceeds in the standard PFN fashion: synthetic SCMs are drawn from a prior $p(\psi)$, paired observational/interventional data are generated, and the model is trained by cross-entropy on the correct outcome bin.

4 Proposed direction

We propose three architectural/output-side changes (Sections 4.1–4.4), one inference-time conditioning component (Section 4.6), and one training-time prior modification (Section 4.7). These components are independent and can be adopted separately. The overall structure is shown in Figure 2.

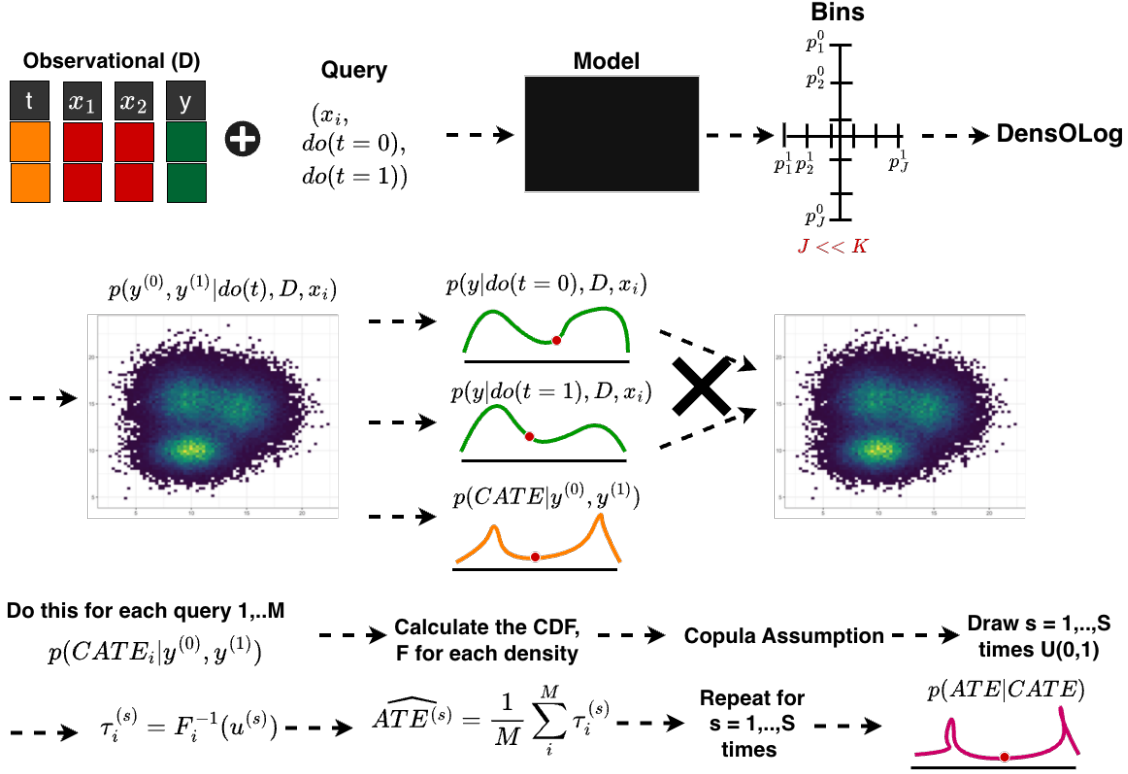


Figure 2: The CFM for Decision-Makers. The model outputs a two-dimensional bar distribution over the paired potential outcomes ($Y^{do(1)}, Y^{do(0)}$) on a $J \times J$ grid, with $J \ll K$ so the joint head remains computationally light. Marginal interventional distributions are recovered by integrating out one axis; the CATE distribution is obtained by aggregating bin pairs whose centre differences fall in the same τ -bin. DensOLog density refinement is applied to the coarse $J \times J$ output to recover smooth joint, marginal, and CATE densities.

4.1 Paired potential-outcome head

In the Do-PFN formulation [1], the model approximates the posterior predictive over a single intervention, marginalised over the (implicit) SCM posterior:

$$p(Y | D_{\text{obs}}, x, do(t)) = \int p(Y | x, \psi_{do(t)}) p(\psi | D_{\text{obs}}) d\psi. \quad (1)$$

The corresponding model output is $q_{\theta}(Y | D_{\text{obs}}, x, do(t))$, trained to approximate Equation (1) via the forward-KL PFN objective on synthetic data.

We replace this with the analogous quantity over the *paired* potential outcomes:

$$p(Y^{do(1)}, Y^{do(0)} | D_{\text{obs}}, x) = \int p(Y^{do(1)}, Y^{do(0)} | x, \psi) p(\psi | D_{\text{obs}}) d\psi. \quad (2)$$

The corresponding model output is $q_\theta(Y^{do(1)}, Y^{do(0)} \mid D_{\text{obs}}, x, do(1), do(0))$, trained to approximate Equation (2).

Why the joint does not factorise. Within a single SCM $\psi = (\mathcal{G}, \{f_k\}, p(\varepsilon))$, the two potential outcomes are generated by intervening on the same structural equations and propagating through the *same* exogenous noise draw:

$$p(Y^{do(1)}, Y^{do(0)} \mid x, \psi) = \int p(Y^{do(1)} \mid x, \psi_{do(1)}, \varepsilon) p(Y^{do(0)} \mid x, \psi_{do(0)}, \varepsilon) p(\varepsilon) d\varepsilon. \quad (3)$$

The shared ε is what couples $Y^{do(1)}$ and $Y^{do(0)}$. Consequently Equation (2) does *not* equal the product of the two marginal posterior predictives,

$$p(Y^{do(1)}, Y^{do(0)} \mid D_{\text{obs}}, x) \neq p(Y^{do(1)} \mid D_{\text{obs}}, x, do(1)) p(Y^{do(0)} \mid D_{\text{obs}}, x, do(0)), \quad (4)$$

which is the failure mode used by the counter-example in Section 2: convolving the two independent marginals introduces values of τ that pair ψ_a 's treated outcome with ψ_b 's control outcome, even though no single ψ in the posterior would produce that pairing.

Recovering the marginals. Integrating one axis of Equation (2) recovers the Do-PFN marginal of Equation (1):

$$p(Y^{do(t)} \mid D_{\text{obs}}, x, do(t)) = \int p(Y^{do(1)}, Y^{do(0)} \mid D_{\text{obs}}, x) dY^{do(1-t)}, \quad t \in \{0, 1\}. \quad (5)$$

So no information modelled by Do-PFN is lost; the joint contains strictly more, namely the coupling between the two potential outcomes.

CATE distribution. The CATE distribution is the induced distribution of $\tau(x) = Y^{do(1)} - Y^{do(0)}$:

$$p(\tau \mid D_{\text{obs}}, x) = \int p(y + \tau, y \mid D_{\text{obs}}, x) dy. \quad (6)$$

Replacing the joint with the product of marginals in this integral is what gives the spurious convolution modes; using the true joint of Equation (2) eliminates them.

Training objective. The model is trained by the standard PFN forward-KL objective on the joint, analogous to the Do-PFN training loss [1]:

$$\arg \min_{\theta} \mathbb{E}_{p(D_{\text{obs}}, x, Y^{do(1)}, Y^{do(0)})} [-\log q_\theta(Y^{do(1)}, Y^{do(0)} \mid D_{\text{obs}}, x)], \quad (7)$$

which is equivalent to minimising $\mathbb{E}_{p(D_{\text{obs}}, x)} [\text{KL}(p \parallel q_\theta)]$ between Equation (2) and q_θ . The PFN training expectation is taken over synthetic SCMs $\psi \sim p(\psi)$, with paired potential outcomes obtained by propagating a *single* draw of ε through both $\psi_{do(1)}$ and $\psi_{do(0)}$, exactly the coupling structure of Equation (3).

In the $J \times J$ bar implementation, this forward-KL objective reduces to cross-entropy on the joint bin label, as described in Section 4.2.

4.2 Two-dimensional bar implementation with $J \ll K$

A direct implementation of the paired potential-outcome head is a two-dimensional bar distribution. Let $B_a \times B_b$, $a, b = 1, \dots, J$, denote a uniform two-dimensional partition of the paired outcome space for $(Y^{do(1)}, Y^{do(0)})$. For a query covariate value x , the neural output head produces a $J \times J$ probability table

$$P_{ab}(x) = q_\theta((Y^{do(1)}, Y^{do(0)}) \in B_a \times B_b \mid D_{\text{obs}}, x, do(1), do(0)), \quad a, b = 1, \dots, J,$$

with

$$P_{ab}(x) \geq 0, \quad \sum_{a=1}^J \sum_{b=1}^J P_{ab}(x) = 1.$$

This table is a coarse binned representation of the joint paired potential-outcome distribution. Its role is to preserve the dependence between $Y^{do(1)}$ and $Y^{do(0)}$, rather than predicting the two interventional outcome distributions independently.

During synthetic pretraining, the paired potential outcomes are generated under the same SCM, propagating a single draw of the exogenous noise through both $\psi_{do(1)}$ and $\psi_{do(0)}$ (Equation (3)). If the ground-truth pair for query x_i falls in cell (a_i, b_i) , the head is trained with the joint cross-entropy loss

$$\mathcal{L}_{\text{joint}} = -\log P_{a_i b_i}(x_i).$$

Thus the neural model learns where the paired potential-outcome mass lies on a coarse two-dimensional grid.

Why $J \ll K$. A direct port of a standard K -bin one-dimensional bar distribution to two dimensions would require K^2 joint cells. For $K = 1000$ this is 10^6 cells per query, which is impractical as an output head and produces a sparse cross-entropy signal. We therefore use a much smaller per-axis resolution J , yielding J^2 joint cells; for example $J = 50$ gives 2,500 cells and $J = 100$ gives 10,000. The constraint

$$J \ll K$$

keeps the J^2 output head comparable in size to a standard K -bin marginal head.

4.3 DensOLog refinement and derived causal distributions

The coarse $J \times J$ table is not used as the final estimate. For each query x , the binned probabilities $\{P_{ab}(x)\}_{a,b=1}^J$ are passed to DensOLog [8], a log-concave density-recovery method. DensOLog has a bivariate version, which recovers a continuous log-concave joint, optionally multimodal, density for the paired potential outcomes directly from the binned input. This step is computationally cheap and takes negligible time.

To obtain a concrete, standardised representation of this continuous density, we evaluate it on a very tight regular grid

$$\{(y_m^{(1)}, y_n^{(0)})\}$$

with grid width δ . This yields the array

$$\hat{q}_x(y_m^{(1)}, y_n^{(0)}) \approx q_\theta(Y^{do(1)} = y_m^{(1)}, Y^{do(0)} = y_n^{(0)} \mid D_{\text{obs}}, x, do(1), do(0)).$$

All reported one-dimensional quantities are derived from the tight-grid joint $\hat{q}_x$ by summation along grid axes, with grid width δ .

Marginal potential-outcome densities. The treated and control potential-outcome densities are obtained by summing the joint over one grid axis:

$$\hat{q}_x^{(1)}(y_m^{(1)}) = \sum_n \hat{q}_x(y_m^{(1)}, y_n^{(0)}) \delta, \quad \hat{q}_x^{(0)}(y_n^{(0)}) = \sum_m \hat{q}_x(y_m^{(1)}, y_n^{(0)}) \delta.$$

Treatment-effect density. The treatment effect $\tau = y^{(1)} - y^{(0)}$ is a unit-Jacobian linear map of the paired outcomes, so its density is the joint summed along the anti-diagonal. On the shared tight grid, the effect grid $\{t_\ell\}$ is the difference grid of the two axes, and

$$\hat{q}_x^\tau(t_\ell) = \sum_n \hat{q}_x(t_\ell + y_n^{(0)}, y_n^{(0)}) \delta.$$

Because the grid is shared and tight, the coordinates $t_\ell + y_n^{(0)}$ are themselves grid coordinates, so this is a direct sum over stored grid values.

Moments of the treatment effect. DensOLog is equipped with an EM procedure for the bivariate log-concave fit which converges to the true mean and covariance structure of the joint paired potential-outcome distribution; we refer to [8] for the proof. Concretely, the EM procedure returns the joint moments

$$\mathbb{E}[Y^{do(1)}], \quad \mathbb{E}[Y^{do(0)}], \quad \text{Var}[Y^{do(1)}], \quad \text{Var}[Y^{do(0)}], \quad \text{Cov}(Y^{do(1)}, Y^{do(0)}).$$

Because the treatment effect $\tau = Y^{do(1)} - Y^{do(0)}$ is a linear function of the paired outcomes, its first two moments are exact functionals of the joint moments above and require no separate fit on the τ -scale:

$$\begin{aligned} \mathbb{E}[\tau(x)] &= \mathbb{E}[Y^{do(1)}] - \mathbb{E}[Y^{do(0)}], \\ \text{Var}[\tau(x)] &= \text{Var}[Y^{do(1)}] + \text{Var}[Y^{do(0)}] - 2 \text{Cov}(Y^{do(1)}, Y^{do(0)}). \end{aligned}$$

The covariance term is precisely what the paired head preserves and what a marginals-only model cannot supply: under independent marginals it would be dropped, inflating $\text{Var}[\tau(x)]$. The EM procedure thus carries over from the joint to the treatment effect without modification — the treatment-effect mean and variance inherit the convergence guarantee of the joint EM through the linear relationship above.

Why this differs from convolving marginals. The treatment-effect density above is a sum over the *joint* $\hat{q}_x$ along anti-diagonals. Replacing $\hat{q}_x(t_\ell + y_n^{(0)}, y_n^{(0)})$ with the product $\hat{q}_x^{(1)}(t_\ell + y_n^{(0)}) \hat{q}_x^{(0)}(y_n^{(0)})$ recovers the naive convolution of the two marginals — the same algebraic shape, but with the coupling between $Y^{do(1)}$ and $Y^{do(0)}$ discarded. That substitution is what produces the spurious treatment-effect values of the counter-example in Section 2. Summing over the genuine joint $\hat{q}_x$ retains the coupling and eliminates them.

The pipeline is therefore:

$$\text{coarse } J \times J \text{ neural table} \longrightarrow \hat{q}_x \longrightarrow (\hat{q}_x^{(1)}, \hat{q}_x^{(0)}, \hat{q}_x^\tau).$$

The neural model learns joint probability mass on a feasible coarse grid; bivariate DensOLog recovers a continuous log-concave joint density from that binned representation; the tight grid of width δ standardises that continuous density; the marginal potential-outcome densities and the treatment-effect density are obtained by summation along grid axes; and the treatment-effect mean and variance follow from the joint moments returned by the EM procedure.

4.4 Shared-rank aggregation for the ATE distribution

After DensOLog refinement, the paired head gives a per-query CATE distribution with CDF

$$F_i(t) = q_\theta(\tau(x_i) \leq t \mid D_{\text{obs}}, x_i).$$

The ATE point estimate remains the average of CATE means,

$$\widehat{\text{ATE}} = \frac{1}{M} \sum_{i=1}^M \mathbb{E}_\theta[\tau(x_i) \mid D_{\text{obs}}, x_i].$$

For an ATE *distribution*, the pointwise CATE marginals alone do not identify a unique answer, because how the per-individual CATEs co-vary across i determines the spread of the ATE. We adopt a *shared-rank coupling* (comonotonic, driven by a single latent uniform): for $s = 1, \dots, S$,

$$u^{(s)} \sim \text{Uniform}(0, 1), \quad \tau_i^{(s)} = F_i^{-1}(u^{(s)}), \quad \text{ATE}^{(s)} = \frac{1}{M} \sum_{i=1}^M F_i^{-1}(u^{(s)}).$$

The empirical distribution of $\{\text{ATE}^{(s)}\}_{s=1}^S$ is reported as the ATE posterior under shared-rank coupling. Sampling independently per individual would artificially average out the model’s shared epistemic uncertainty over causal mechanisms; the shared-rank construction keeps that “which causal world” uncertainty correlated across queries, which is the conservative and decision-relevant choice. Alternative couplings (independent, Gaussian copula, learned) are discussed in Section 6.

4.5 Shortest $(1 - \alpha)$ region.

We additionally report the shortest subset of the treatment-effect axis carrying probability mass $1 - \alpha$, also called the highest-density region (HDR). For a density $\hat{q}_x^\tau$ at level α , this set is

$$C_{1-\alpha}(x) = \{t : \hat{q}_x^\tau(t) \geq h_\alpha\}, \quad h_\alpha = \sup\left\{h \geq 0 : \int_{\{t: \hat{q}_x^\tau(t) \geq h\}} \hat{q}_x^\tau(t) dt \geq 1 - \alpha\right\}.$$

Equivalently, $C_{1-\alpha}(x)$ is obtained by taking the highest-density points first until the accumulated mass reaches $1 - \alpha$. For unimodal $\hat{q}_x^\tau$ it is a single interval; for multimodal $\hat{q}_x^\tau$ it is a union of disconnected intervals

$$C_{1-\alpha}(x) = [a_1, b_1] \cup [a_2, b_2] \cup \dots,$$

with low-density regions between modes excluded. For uniform $\hat{q}_x^\tau$ the set is non-unique and an additional convention is required.

On the tight effect-grid $\{t_\ell\}$ with width δ , the HDR is computed by sorting $\{\hat{q}_x^\tau(t_\ell)\}$ in decreasing order, accumulating mass in units of δ until the cumulative mass first reaches $1 - \alpha$, and reporting all grid points whose density exceeds the resulting threshold. Consecutive grid points are grouped into intervals. The same construction applied to $\{\text{ATE}^{(s)}\}_{s=1}^S$ from Section 4.4 yields an HDR for the ATE distribution.

4.6 Conditioning on partial ancestral information

We will incorporate the partial-graph conditioning mechanism of “Use What You Know” [2] unchanged at the architecture level: a partially-known ancestral matrix (PAM) $\tilde{T} \in \{-1, 0, 1\}^{(d+2) \times (d+2)}$ is injected into the feature-wise attention via learnable soft attention biases, optionally combined

with a graph convolutional encoder feeding the transformer through adaptive layer normalisation. This component is orthogonal to the paired output head: the soft attention bias acts on the input representation, while our changes act on the output representation.

The two interact constructively. When informative graph information is supplied, the model’s implicit posterior over SCMs concentrates, which should narrow the joint $q_{\theta}(Y^{do(1)}, Y^{do(0)} | \dots)$ and correspondingly sharpen the CATE distribution. When no graph information is supplied, the joint can still be diffuse or multimodal, and the decision-friendly head exposes that uncertainty rather than collapsing it. Both behaviours are appropriate.

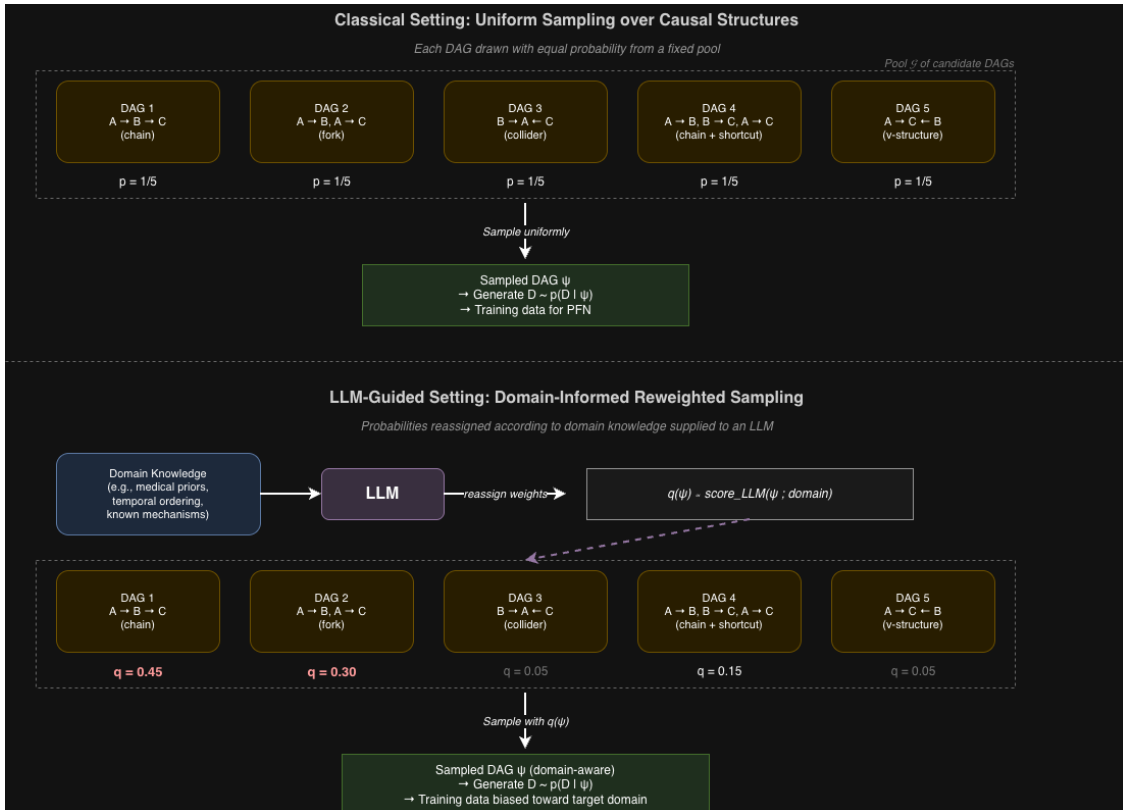
4.7 LLM-guided causal prior reweighting

PFN training samples synthetic SCMs from a fixed prior $p(\psi)$ in which DAGs are typically drawn uniformly (or from a generic Erdős–Rényi distribution). When a target application domain is fixed in advance (e.g. medical, economic), uniformly sampling DAGs spends most training capacity on causally implausible structures for that domain.

We propose an *LLM-guided sampler* that reweights the prior toward domain-relevant causal structures:

$$q(\psi) \propto \text{score}_{\text{LLM}}(\psi; \text{domain}) \cdot p(\psi).$$

The LLM consumes a description of the domain — prior literature, temporal ordering, mechanism plausibility, expert constraints — and assigns a score to each candidate DAG. Sampling proceeds from $q(\psi)$ rather than $p(\psi)$; equivalently, one can implement this as importance reweighting of training batches. This component is orthogonal to the output head: it changes the synthetic training distribution, not the architecture. See Figure 4.7.



5 Empirical plan

A direct evaluation of the recovered CATE *distribution* against ground truth is not possible: real datasets only provide ground-truth point values of potential outcomes, not ground-truth posteriors over causal effects. We therefore would need a separate evaluation metric that combines standard point-estimate metrics.

- (1) **Marginal calibration is preserved.** Confirm that the paired head does not degrade marginal predictive performance on Y . Compare 1D NLL on held-out $(Y, x, do(t))$ triples between (a) the standard marginal Do-PFN head and (b) the marginals obtained by integrating our $J \times J$ joint. The latter should match or beat the former.
- (2) **Standard causal benchmarks.** Evaluate $\sqrt{\text{PEHE}}$ and ε_{ATE} on RealCause [7] (IHDP, ACIC, CPS, PSID), directly comparing against the numbers in “Use What You Know” [2]. Goal: match or improve point-estimate performance while *additionally* producing the distributional outputs. Probably considering different metrics as well!
- (3) **J /DensOLog ablation.** Sweep $J \in \{20, 50, 100, 200\}$ with and without DensOLog refinement. This is for optimal performance/cost choice of J .
- (4) **Naive-convolution comparison.** On synthetic SCMs from the training prior, compare the τ -distribution recovered from our joint head against the τ -distribution obtained by independently convolving the two marginal heads. Quantify the discrepancy and the prevalence of spurious modes in the convolution baseline.
- (5) **LLM-guided prior ablation.** For a domain-tagged benchmark (e.g. IHDP with its medical context), compare models trained under the uniform prior versus the LLM-reweighted prior. Measure whether the reweighted prior reduces $\sqrt{\text{PEHE}}$ and sharpens the recovered CATE distribution.
- (6) **Qualitative decision-maker case studies.** Identify concrete settings on RealCause where the marginal-mean pipeline produces a misleadingly confident scalar (e.g. a CATE point estimate near zero) while the paired head exposes substantial spread in τ . These case studies are the primary evidence for the decision-friendly framing.

6 Discussion and open questions

Goal. A CFM whose output is a distribution over treatment effects rather than a point estimate, faithfully exposing the model’s own epistemic uncertainty for the treatment effect and NOT interventional distribution of Y to the decision-maker, and sharpening that distribution when partial causal information is supplied at test time.

Open questions.

- *Coupling for ATE.* Shared-rank coupling is one principled choice. Others (independent, Gaussian copula, learned via the model) should be compared empirically and theoretically; the right default may depend on the application.
- *Extending beyond binary treatment.* It is possible but as a next paper.
- *Validating the LLM-guided prior.* Reweighting the training prior using an LLM raises the question of whether the resulting distribution reflects true domain structure or LLM bias. Held-out domain benchmarks and sensitivity analyses are needed.

- *Calibration of the joint.* Diagnostics for whether $q_\theta(Y^{do(1)}, Y^{do(0)} \mid \dots)$ is well-calibrated as a joint — not just as marginals — need development; existing calibration tools are mostly 1D.

Contribution summary. The proposal contributes:

- a critique of the marginals-only output of existing CFMs and its consequences for decision-making, illustrated by a counter-example in which naive aggregation invents spurious modes;
- a paired potential-outcome head implemented as a $J \times J$ bar distribution with $J \ll K$, trained by cross-entropy on the joint bin label, which preserves the coupling between $Y^{do(1)}$ and $Y^{do(0)}$;
- DensOLog density refinement on the coarse joint output, decoupling bin count from density quality;
- shared-rank aggregation of pointwise CATE distributions into an ATE distribution that preserves dependence across queries;
- an LLM-guided reweighting of the synthetic training prior toward domain-relevant DAGs.

7 Appendix

7.1 Mean Recovery - From DensOLog Paper

A key component of our work is to first estimate the true mean of the unobserved data. We do this by assuming that the data follow a Gaussian distribution and applying the EM algorithm, as in [CITE]. Although Gaussian densities are log-concave, we expect the Gaussian assumption to be incorrect in general. Nevertheless, the motivation for using this approach is that, under the Kullback–Leibler (KL) divergence, the Gaussian density g closest to a fixed density f_0 is the Gaussian density with mean $\mu_0 = \int x f_0(x) dx$. Recall that the KL divergence is defined as

$$\rho_{KL}(g|f_0) = \int f_0 \log f_0 dx - \int f_0 \log g dx.$$

For a log-concave density, the first term $\int f_0 \log f_0 dx$ exists. Therefore, minimizing the KL divergence over some class of densities $g \in \mathcal{G}$ is equivalent to minimizing the second term. Taking $g = g_\mu$ to be a Gaussian density with mean μ and fixed variance $\sigma^2 > 0$, we can calculate

$$-2 \int f_0(x) \log g_\mu(x) dx = -\frac{1}{\sigma^2} \int (x - \mu)^2 f_0(x) dx - \log \sigma^2 + c, \quad (8)$$

for some constant c . Minimizing this expression with respect to μ yields the result. We assume that the variance is known and therefore suppress its dependence in the notation. Because the data are grouped, the above relationship no longer holds exactly, even asymptotically. However, the error due to grouping is quantifiable as we demonstrate in what follows.

The log-likelihood for the grouped data under IID sampling and assuming a Gaussian density is given by

$$\ell_n(\mu) = \sum_{j=1}^k p_{n,j} \log \int_{a_j}^{a_{j+1}} g_\mu(x) dx. \quad (9)$$

Note also the mild abuse of notation as the log-likelihood depends on the data (via the empirical weights $\{p_{n,j}\}$) as well as the pre-defined grid $\{a_j\}$. The partial derivatives of $\ell_n(\mu)$ can be calculated easily as

$$\partial_\mu \ell_n(\mu) = \sum_{j=1}^k p_{n,j} \frac{\int_{a_j}^{a_{j+1}} \frac{x-\mu}{\sigma} g_\mu(x) dx}{\sigma \int_{a_j}^{a_{j+1}} g_\mu(x) dx} = \sum_{j=1}^k p_{n,j} \frac{\int_{\alpha_j}^{\alpha_{j+1}} u \phi(x) dx}{\int_{\alpha_j}^{\alpha_{j+1}} \phi(x) dx}$$

where $\alpha_j = (a_j - \mu)/\sigma$ for $j = 1, \dots, k+1$. We also use ϕ to denote the density of the standard normal random variable. Let f_0 denote the true log-concave density of the unobserved data $X_1, \dots, X_n$. Our first result is the following.

Theorem 1. *Assume that $\delta < 2\sigma$. The maximizer $\hat{\mu}_n$ of the log-likelihood (9) exists and is unique. Furthermore, as $n \rightarrow \infty$, $\hat{\mu}_n \rightarrow \hat{\mu}_0$, where $\hat{\mu}_0$ is the unique solution to the equation*

$$s_0(\mu) \equiv \sum_{j=1}^k p_{0,j} \frac{\int_{a_j}^{a_{j+1}} \frac{x-\mu}{\sigma} g_\mu(x) dx}{\int_{a_j}^{a_{j+1}} g_\mu(x) dx} = \sum_{j=1}^k p_{0,j} \frac{\int_{\alpha_j}^{\alpha_{j+1}} u \phi(x) dx}{\sigma \int_{\alpha_j}^{\alpha_{j+1}} \phi(x) dx} = 0,$$

where $p_{0,j} = \int_{a_j}^{a_{j+1}} f_0(x) dx$.

From (9), we obtain the (modified) score function

$$s_n(\mu) = \partial_\mu \ell(\mu).$$

It is possible to verify that the equation in the above theorem is simply the score function used to minimize the Kullback-Liebler divergence between the probability mass function $p_{0,j}$ and $\int_{a_j}^{a_{j+1}} g_\mu(x) dx$.

Theorem 2. *Let $a_j, j = 1, \dots, k+1$ denote a uniform partition with $\delta = a_{j+1} - a_j$, for all j . Then, as $\delta \rightarrow 0$, we have that $\hat{\mu}_0 \rightarrow \mu_0$. Furthermore, for a fixed uniform partition, we have that there exists an $N = N(\omega)$, such that for all $n \geq N(\omega)$, $|\mu_0 - \hat{\mu}_n| \leq \delta$. Furthermore,*

$$|\mu_0 - \hat{\mu}_0| \leq \delta.$$

This last result is key to our mean recovery method. We will use the EM algorithm (described in detail in Appendix ??) to find the MLE associated with (9). The two theorems above state that this MLE exists and is unique, and converges to the true population mean (without grouping) as the sample size grows and the grouping becomes more refined.

Estimated mean would be theoretically at most 0.02 different from true mean for $J = 100$ and 0.04 for $J = 50$

7.2 Consistency and compatibility - From DensOLog Paper

We next examine the large sample behaviour of our proposed estimator, both for the cases when $n \rightarrow \infty$ as well as $\delta \rightarrow 0$.

The following statement follows directly from the results of JankowskiDiscr2013.

Lemma 3. *Consider the probability mass function $\{p_{0,j}\}$. Define $\hat{p}_0$ as $\hat{p}_0 = e^{\hat{\varphi}_0}$, where $\hat{\varphi}_0$ is the discretely concave function which maximizes*

$$\sum_j \varphi_{0,j} p_{0,j} - \sum_j e^{\varphi_{0,j}} p_{0,j} + 1.$$

Then $\hat{p}_0$ exists and is unique, and satisfies $\sum_j \hat{p}_{0,j} a_j = \sum_j p_{0,j} a_j$. Moreover, if p_0 is discretely log-concave, then $\hat{p}_0 = p_0$. Furthermore, as $n \rightarrow \infty$, $\hat{p}_n \rightarrow \hat{p}_0$, almost surely.

Next, define $G_{n,\delta}$ to be the cumulative distribution function of the random variable $Z_{n,\delta}$. We assume that there exists a random variable Z_δ with cumulative distribution function G_δ , such that

$$\int_0^\delta |\tilde{G}_{n,\delta} - \tilde{G}_\delta|(x)dx \rightarrow 0 \quad (10)$$

as $n \rightarrow \infty$. Note that this is convergence in Mallows' distance. From [Lemma 8.3]bickel84, this implies that $\hat{\mu}_n - \sum_{j=1}^k p_{n,j} a_j = E[Z_{n,\delta}] \rightarrow E[Z_\delta]$. Therefore, $E[Z_\delta] = \hat{\mu}_0 - \sum_j a_j p_{0,j}$, from Theorem 2. Note also that our specific choice of $Z_{n,\delta}$ in the previous section satisfies condition (10). Indeed, the distribution of G_δ is δZ , where Z is a beta random variable with α as originally chosen by the user and

$$\beta = \alpha \left(\frac{\delta}{\hat{\mu}_0 - \sum_j a_j p_{0,j}} - 1 \right).$$

Finally, let $Q_{n,\delta}$ denote the distribution of the convolution of the distributions $\hat{p}_n$ and $G_{n,\delta}$ (this is also the distribution of $X_{n,\delta}$ as defined in the previous section). Similarly, let Q_δ denote the convolution of $\hat{p}_0$ and G_δ .

Theorem 4. *Let $\hat{f}_0$ denote the closest log-concave density to the distribution Q_δ in the sense of Kullback-Liebler divergence. That is, the density $\hat{f}_0$ is the unique log-concave density which minimizes*

$$\int \varphi dQ_\delta - \int e^\varphi dQ_\delta + 1,$$

over all concave functions φ . Denoting the minimizer as $\hat{\varphi}_0$, we have that $\hat{f}_0 = e^{\hat{\varphi}_0}$. With δ fixed, as $n \rightarrow \infty$, the density estimator $\hat{f}_n$ converges to $\hat{f}_0$. That is,

$$\limsup_n \int_{-\infty}^{\infty} |\hat{f}_n - \hat{f}_0|(x)dx = 0.$$

This result relies heavily on [Theorem 2.15]DSS2011, where log-concave projections are shown to be continuous in Mallows' distance. As a corollary, we also obtain consistency as δ decreases.

Corollary 5. *Suppose that the generating density f_0 is log-concave. Then, as $n \rightarrow \infty$ and $\delta \rightarrow 0$, the density estimator $\hat{f}_n$ is consistent. That is,*

$$\limsup_\delta \limsup_n \int_{-\infty}^{\infty} |f_n^* - f_0|(x)dx = 0,$$

almost surely, for $f_n^ = \hat{f}_n$ or $\hat{f}_n^{smooth}$.*

Main comment is regarding the prior. Would you trust it as a patient? Prior!!!!

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